

Requirements Development and Management (RDM)

PA Overview

Required PA Information

Intent

Elicits requirements, ensures common understanding by stakeholders, and aligns requirements, plans, and work products.

Value

Ensures that customers' needs and expectations are satisfied.

Additional Required PA Information

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Explanatory PA Information

Practice Summary

Level 1

RDM 1.1	Record requirements.

Level 2

RDM 2.1	Elicit stakeholder needs, expectations, constraints, and interfaces or connections.
RDM 2.2	Transform stakeholder needs, expectations, constraints, and interfaces or connections into prioritized customer requirements.
RDM 2.3	Develop an understanding with the requirements providers on the meaning of the requirements.
RDM 2.4	Obtain commitment from project participants that they can implement the requirements.
RDM 2.5	Develop, record, and maintain bidirectional traceability among requirements and activities or work products.
RDM 2.6	Ensure that plans and activities or work products remain consistent with requirements.

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Level 3

RDM 3.1	Develop and keep requirements updated for the solution and its components.
RDM 3.2	Develop operational concepts and scenarios.
RDM 3.3	Allocate the requirements to be implemented.
RDM 3.4	Identify, develop, and keep updated interface or connection requirements.
RDM 3.5	Ensure that requirements are necessary and sufficient.
RDM 3.6	Balance stakeholder needs and constraints.
RDM 3.7	Validate requirements to ensure the resulting solution will perform as intended in the target environment.

Additional PA Explanatory Information

Three typical types of requirements:

1. Customer or business requirements
2. Solution requirements
3. Interface or connection requirements

Taken together, these requirements address the needs of stakeholders, including needs pertinent to various lifecycle phases and attributes, e.g., responsiveness, security, quality. Requirements can also include constraints.

All projects have requirements. Requirements are the basis for developing the right solutions. Requirements development activities include:

- Eliciting, analyzing, validating, and communicating customer needs, expectations, and constraints
- Prioritizing customer requirements to understand what will satisfy stakeholders given resource constraints
- Developing the lifecycle requirements of the solution
- Developing operational concepts and scenarios
- Developing the customer functional and quality attribute requirements, including descriptions, decompositions, and allocating requirements to functions
- Developing initial solution requirements consistent with customer requirements

Customer requirements are further refined into solution and interface or connection requirements. In addition to customer requirements, solution and interface or connection requirements are derived from the selected design solutions.

Identify and refine requirements throughout the project. Involve stakeholders in requirements development and analysis activities to give them visibility into the evolution of requirements. Analyze design decisions, subsequent corrective actions, and feedback for impact on requirements. Analyses can be used to understand, define, and select the requirements.

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In addition, the definition may specify design constraints. Some quality attributes will emerge as architecturally significant and drive the development of the solution architecture. Quality attributes may address:

- Solution availability
- Ability to sustain and maintain
- Timeliness, throughput, and responsiveness
- Consistency
- Security
- Scalability

Analyses are iterated until there is enough detail to develop the solution or a portion of the solution. Analysis of requirements and the operational concepts and scenarios may result in identifying more requirements, including:

- Constraints of various types
- Technological limitations
- Costs
- Time constraints
- Risks
- Functionality, support, and maintenance concerns
- Issues implied but not explicitly stated by the customer
- Business considerations, regulations, and laws

Develop a functional design through iteration with the evolving operational concept and scenarios. During design, refine, derive, and allocate requirements to the functional solution and solution components.

Related Practice Areas

Peer Reviews (PR)

Verification and Validation (VV)

Context Specific

Agile with Scrum Guidance

Context Tag:	Agile with Scrum
Context:	Integrate agile techniques and ceremonies with other processes.

Figure RDM-1: RDM in an Agile Framework states where requirements activities are performed in an agile project using Scrum. Figure RDM-2: Requirements Information shows a top-level summary of requirements information and in which sprint those requirements are addressed.

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Figure RDM-1: RDM in an Agile Framework

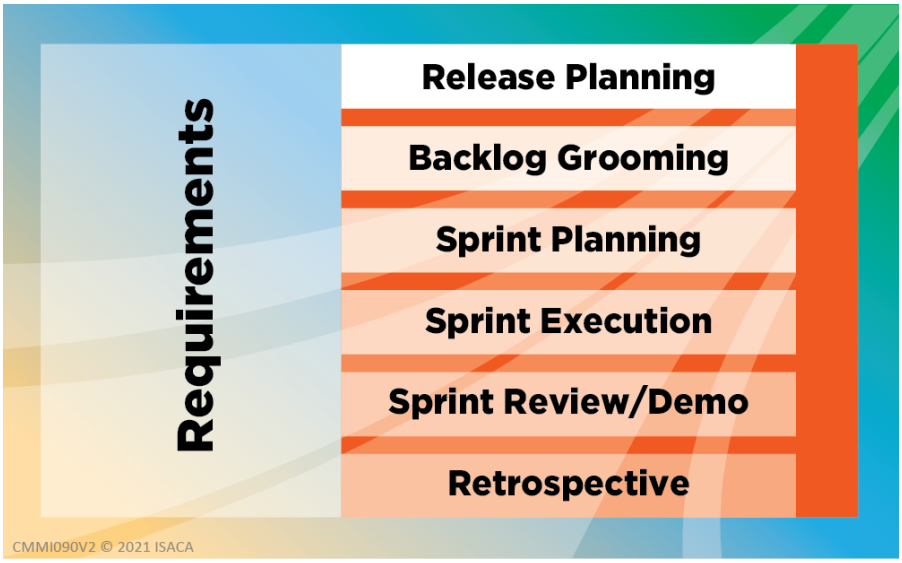


Figure RDM-2: Requirements Information



Requirements Development and Management activities can strengthen the typical agile project by adding greater requirements understanding, clarity, and discovery of issues. These activities add early consideration of other requirement types beyond the user stories typically used by an agile project using Scrum. These activities also include performing analyses to find errors and risks while conducting the requirements definition activities. For example, agile expects user needs to be elicited as a backlog of user stories, but a backlog does not typically include constraints, interfaces, and quality attributes. RDM practices provide a means for capturing and addressing these attributes during the sprints. Requirements Development and Management practices provide a robust infrastructure to support requirements development for complex solutions.

These practices can be added iteratively to improve any agile project using Scrum during backlog creation, grooming, and sprint execution.

Table RDM-1: Scrum Examples shows where Requirements Development and Management practices can augment a typical agile project using Scrum.

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Table RDM-1: Scrum Examples

Agile Projects using Scrum	Requirements Development and Management
Release planning	Earlier and more complete understanding of the solution and risks.
Backlog grooming/review	Broader and deeper analysis of user stories or epics to discover potential issues or constraints. Additionally, these analyses will identify risks or opportunities.
Sprint planning	Presentation of stories by the product owner, review acceptance by the team, and estimate of the user stories to be delivered during the upcoming sprint.
Sprint execution	More of the project is spent developing a working solution vs. refactoring.
Sprint review/demo	Enables a more thorough understanding of what has been accomplished during the sprint.
Sprint retrospective	Collaborative sessions where the agile team's culture, process, and performance is reviewed

Agile projects using Scrum typically will implement traceability from business need through epics, user stories, tasks, tests, and the definition of done. Designs and code are often traced directly to user stories. Traceability enables more efficient and accurate consistency checks between requirements (user stories or epics) and work products. Traceability also improves the ability to understand and addresses what is impacted by a requirement (user story or epic) change.

Development

Context Tag:	CMMI-DEV
Context:	Use processes to develop quality products and services to meet the needs of customers and end users.

For product lines, engineering processes (including requirements development) may be applied at multiple levels. At the product line level, perform a "commonality and variation analysis" to help elicit, analyze, and develop core assets for use by projects within the product line. At the project level, use these core assets per the product line plan as part of the project's engineering activities.

Security

Context Tag:	CMMI-SEC
Context:	Use processes to incorporate security considerations as an integral part of the solution, work, project, and organization.

Determine security needs by addressing and analyzing inputs from various sources, as derived from:

- Security needs of relevant stakeholders
- Results of security and threat assessments

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- Threat intelligence analysis

Collect inputs for security requirements from stakeholders, consolidate inputs, analyze the intended operational environments, and resolve conflicts.

Security sources may include:

- Country or region-specific regulations and laws, e.g., personal data protection, export controls, General Data Protection Regulation (GDPR)
- Request for Proposal (RFP), solicitation, or feature requests
- Solution security risk assessments
- Development of misuse and abuse use cases
- Organization internal protection goals, e.g., intellectual property protection and licensing regulations
- Security activities of competitors
- Analysis of existing or future threats by using vulnerability databases, e.g., Common Weakness Enumeration List; and SysAdmin, Audit, Network, Security (SANS) TOP 25 Most Dangerous Software Errors
- Security topics with high customer profile
- Analysis of security incidents in own or comparable systems
- Organizational security baselines and policies

Refer to Table RDM-2: Security External References for more information.

Table RDM-2: Security External References

External Reference Item	Link
Cybersecurity Standards of the North American Electric Reliability Corporation [NERC 2011 series]	https://www.nerc.com/pa/Stand
Health Insurance Portability and Accountability Act (HIPAA) [US 1996]	https://www.hhs.gov/hipaa
Health Information Technology for Economic and Clinical Health Act (HITECH) [US 2009]	https://www.asha.org/Practice/reimbursement/hipaa/HITECH-Act
White Paper Requirements for Secure Control and Telecommunication Systems [BDEW 2008]	https://www.ids.de/fileadmin/Presse/Publikationen/conrad_michael.pdf
Embedded Device Security Assurance Certification Specification [ISCI 2010]	https://www.isasecure.org/en-US/News-Events/ISCI-Publishes-ISASecure-EDSA-Certification-Specif
Security Requirements for Vendors of the International Instrument Users' Association WIB [WIB 2010]	http://osgug.ucaiug.org/conformity/security/Shared%20Documents/WIB%20M2784%20PCS%20VendorSecurity%20v2.pdf

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External Reference Item	Link
System security requirements and security assurance levels of the ISA99 Committee [ISA 2011]	https://www.isa.org/isa99
Common Criteria for Information Technology Security Evaluation [CCMB 2009]	https://www.commoncriteriaportal.org/files/ccfiles/CCPART1V3.1R3.pdf
PCI Security Standards Council (SSC) Data Security Standard [SSC 2010]	https://www.pcisecuritystandards.org/document_library?category=pcidss&subcategory=pcidss_supporting#results

Supplier Management

Context Tag:	CMMI-SPM
Context:	Use processes to identify, select, and manage suppliers and their agreements.

The acquirer has overall responsibility for ensuring that requirements meet the objectives for the solution. The acquirer should clearly define requirements that can be incorporated into supplier agreements and solutions. In some acquisitions, the acquirer assumes the overall role of engineer, architect, or integrator for the solution.

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Level 1

RDM 1.1

Required Practice Information

Practice Statement

Record requirements.

Value

Addresses customer needs and expectations.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Typically, customers cannot describe all aspects of what they want and need. Recorded requirements provide a basis for mutual discussion, understanding, and agreement between the customer and the project. Requirements include what the delivered solution will do.

Example Activities

Example Activities	Further Explanation
Record the requirements.	

Example Work Products

Example Work Products	Further Explanation
Recorded requirements	Examples include: • List of requirements • Statement of work • Use cases

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Level 2

RDM 2.1

Required Practice Information

Practice Statement

Elicit stakeholder needs, expectations, constraints, and interfaces or connections.

Value

Ensures a deeper mutual understanding of the requirements and increases the likelihood that the customer will be satisfied.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Ensure stakeholder needs, expectations, constraints, limitations, and interfaces or connections are clearly identified, understood, and do not conflict. Determine additional requirements to address lifecycle activities and their effect on the solutions. Use an iterative process throughout the life of the project to continuously refine the requirements. Consider environmental, legal, and other constraints when developing customer requirements. Requirements record externally observable behavior. Recorded internal behavior is a design constraint. Requirements represent what the customer needs and expects, not how the requirements will be addressed.

Example Activities

Example Activities	Further Explanation
Elicit stakeholder needs, expectations, constraints, and interfaces or connections.	Identify additional requirements not explicitly provided by stakeholders. Techniques to elicit needs include: • Technology demonstrations • Interim project reviews • Questionnaires • Interviews • Scenarios (operational, sustainment, and development) • Walkthroughs • Quality attribute elicitation workshops with stakeholders • Prototypes and models • Brainstorming • Quality function deployment • Market surveys • Beta testing

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Example Activities	Further Explanation
	• Extraction from sources such as documents, standards, or specifications • Observation of existing solutions, environments, and workflow patterns • Use cases • Business case analysis • Reverse engineering (for legacy solutions) • Customer satisfaction surveys • Viewpoint analysis

Example Work Products

Example Work Products	Further Explanation
List of stakeholder needs, expectations, constraints	
List of interfaces or connections	Interfaces or connections may include: • Links • System • Human • Relationships • Interactions • Interdependencies

RDM 2.2

Required Practice Information

Practice Statement

Transform stakeholder needs, expectations, constraints, and interfaces or connections into prioritized customer requirements.

Value

Ensures customer priorities are addressed to minimize the cost of rework during acceptance and maximize customer satisfaction.

Additional Required Information

If Safety is included in the selected view: Consider, define, and prioritize product or service system safety requirements throughout the solution lifecycle for any systems, new development, upgrades, modifications, or resolution of incidents and deficiencies. Identify, define, and flow down solution safety requirements to the supply chain, as appropriate.

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If Security is included in the selected view: Consider, define, and prioritize solution security requirements throughout the solution lifecycle for any supporting systems, new development, upgrades, modifications, or resolution of incidents and deficiencies. Identify, define, and flow down solution security requirements to the supply chain, as appropriate.

Explanatory Practice Information

Additional Explanatory Information

Consolidate and prioritize various inputs from customers and stakeholders, obtain missing information, and resolve conflicts as customer requirements are developed, prioritized, and recorded.

The customers' functional and quality attribute requirements can be expressed in the customer's terms and can contain non-technical descriptions. Solution and contractual requirements are the expression of these requirements in more explicit technical terms that can be used for design decisions.

Sources for requirements include:

- Customer provided input
- Stakeholder provided input
- Previous efforts
- Existing solution systems
- Domain literature
- Laws and regulations
- Standards
- Business policies
- Previous architectural design decisions and principles
- Business environmental requirements, e.g., laboratories, testing and other facilities, information technology infrastructure
- Technology

These stakeholder needs, expectations, constraint, and interface or connection requirements may include:

- Technical requirements, such as:
  - External interface or connection
  - Internal interface or connection (developed during design)
  - Functional
  - Quality
  - Operational
  - Performance
  - Verification

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    - Validation
    - Acceptance criteria
    - Safety
    - Security
- Non-technical requirements, including:
    - Price and cost
    - Delivery constraints
    - Resource constraints
    - Training
    - Customer interactions, e.g., status reporting, meetings

Ensure technical and non-technical requirements address the satisfaction of customer, business, and project objectives and associated attributes, such as effectiveness and affordability.

Analyze stakeholder requirements to lay the foundation for the operational concept. To avoid scope creep, develop criteria to designate appropriate channels or official sources from which to receive requirements changes.

This results in more detailed and precise sets of requirements called "derived requirements". These requirements address all aspects of the deliverables including:

- Work products
- Services
- Processes
- Consumables
- Customer-provided resources and other resources
- Functionality and quality attribute needs of affected stakeholders

Derived requirements arise from:

- Constraints
- Consideration of issues implied but not explicitly stated in the stakeholder requirements
- Factors introduced by the selected:
    - Unique business considerations
    - Strategic priorities
    - Industry market and technology trends
    - Architecture
    - Design

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Example Activities

Example Activities	Further Explanation
Translate stakeholder needs, expectations, constraints, and interfaces or connections into recorded customer requirements.	Factors to consider when expressing customer requirements include: • Key characteristics of the desired capability • Obstacles to overcome to achieve the capability • Competitive gap between the existing and the desired capability • Supportability of the desired capability Requirements should not specify or constrain design decisions without careful consideration.
Develop, record, and keep updated a prioritization of customer requirements.	Prioritized customer requirements help to determine project, iteration, or increment scope. This prioritization ensures that functional and quality attribute requirements critical to the customer and other stakeholders are given the highest visibility and attention.

Example Work Products

Work Products	Further Explanation
Prioritized customer requirements	
Customer constraints	May include constraints related to: • Design • Verification • Validation

Context Specific

Safety

Context Tag:	CMMI-SAF
Context:	Use processes to incorporate safety considerations as an integral part of the solution, work, project, and organization.

Requirements help identify and define all known hazards and their associated risks and enable the elimination or reduction of safety risks to acceptable levels. Consider outcomes from safety evaluation activities when identifying safety requirements. Establish traceability of safety requirements to related hazards. Set a safety target for each safety requirement. For example, the safety target should ensure the product's risk is less than or equal to the acceptable risk, based on the minimum acceptable safety tolerance limits. Ensure the safety target and product risk information are carried forward into product design. Identify specific system safety engineering requirements in the system specification including risk assessment and mitigation, fault tolerance, acceptance, unique classifications and certifications, or any unique mishap reduction needs, e.g., detection, isolation, annunciation, and recovery.

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RDM 2.3

Required Practice Information

Practice Statement

Develop an understanding with the requirements providers on the meaning of the requirements.

Value

Ensures the correct solution is delivered which increases customer satisfaction.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Ensure people who implement and test solutions against the requirements analyze them with the provider to reach a shared understanding of the meaning of the requirements. The result of these analyses and interactions is a set of approved requirements. It is important to identify and eliminate as much ambiguity as possible from the set of requirements.

Example Activities

Example Activities	Further Explanation
Develop criteria for identifying appropriate requirements providers.	
Develop criteria for the evaluation and acceptance of requirements.	Evaluation and acceptance criteria can prevent: • Inadequate verification • Costly rework • Customer rejection For example, evaluation and acceptance criteria may include: • Unambiguous • Clearly and properly stated • Complete • Consistent with one another • Uniquely identified • Consistent with the architectural approach and quality attribute priorities • Appropriate to implement • Testable • Traceable to source • Achievable • Tied to business value

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Example Activities	Further Explanation
	• Identified as a priority by the customer
Analyze requirements to ensure that established criteria are met.	
Reach an understanding of and obtain commitments to requirements with the requirements providers and the project participants.	
Record needed changes to requirements.	

Example Work Products

Example Work Products	Further Explanation
Lists of appropriate requirements providers	
Criteria for evaluation and acceptance of requirements	
Results of analyses against criteria	
Recorded changes to requirements	
Set of approved requirements	Reflects the shared understanding between the project and the customers.

Context Specific

Safety

Context Tag:	CMMI-SAF
Context:	Use processes to incorporate safety considerations as an integral part of the solution, work, project, and organization.

Analyze requirements from the perspective of safety considerations. For example, the analysis may include a review of subsystem interrelationships for:

- Compliance with specified Environment, Safety, and Occupational Health (ESOH) design criteria
- Possible independent, dependent, and simultaneous hazardous events, including system failures, failures of ESOH devices, common cause failures and events, and system interactions that could create a hazard or increase risk
- Degradation of a subsystem or the total system affecting ESOH
- Design changes affecting subsystems
- Effects of reasonable human errors

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- The method of implementing hardware, software, and facilities/installation design requirements and corrective actions that have not introduced any new hazards or negatively impacted ESOH-related aspects of the system
- Control of Substances Hazardous to Health (COSHH)

RDM 2.4

Required Practice Information

Practice Statement

Obtain commitment from project participants that they can implement the requirements.

Value

Ensures commitments are well understood to minimize delays and rework.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

As requirements are developed or evolved, ensure project participants commit to the approved requirements and the resulting changes in project plans, activities, and work products. Their commitment increases the likelihood of success in meeting project objectives.

Example Activities

Example Activities	Further Explanation
Assess the impact of requirements on existing commitments.	
Negotiate and record commitments.	Negotiate commitments before work starts

Example Work Products

Example Work Products	Further Explanation
Impact assessment	Estimate the cost, and the impact to quality, risk, and schedule.
Recorded commitments that requirements can be met	Recorded commitments may be in the form of: • Meeting minutes • Sign-off documents • Email acknowledgements • May address resources and schedule.

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Context Specific

Safety

Context Tag:	CMMI-SAF
Context:	Use processes to incorporate safety considerations as an integral part of the solution, work, project, and organization.

Perform a safety impact analysis at any point within the solution lifecycle when requirements change. Some safety groups and standards call these impact analyses System Hazard Analysis (SHA). They are used to verify system compliance with Environment, Safety, and Occupational Health (ESOH) requirements contained in system specifications and other relevant documents.

The impact analysis report includes system information for the solution, with hazard analysis results incorporated, as appropriate. Contents and formats of the hazard information may vary according to the individual requirements of the program. The detailed content for hazard analysis results may include:

- Summary of the analysis results
- A listing of identified hazards, such as:
  - Lifecycle phases affected by the hazard
  - Causal factor, e.g., hardware, software, and human
  - Effects of the hazard
  - Initial risk category and associated attributes
  - Mitigation strategies
  - Hazard status, e.g., open or closed
  - Hazard traceability (running history of actions taken or planned with rationale to mitigate and risks)

Supplier Management

Context Tag:	CMMI-SPM
Context:	Use processes to identify, select, and manage suppliers and their agreements.

The acquirer negotiates with the customer and supplier before committing to a requirement change. A requirements change can lead to modifications to supplier agreements. Ensure the acquirer, supplier, and customer agree on these changes after appropriate negotiations. In some acquisitions, the acquirer may represent, and act on behalf of the customer. Deliverables may include impact assessments when a requirement change occurs.

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RDM 2.5

Required Practice Information

Practice Statement

Develop, record, and maintain bidirectional traceability among requirements and activities or work products.

Value

Ensures consistency between requirements and the solution which increases the likelihood of customer satisfaction.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Traceability from source requirements to the solution ensures that:

- Requirements have been completely addressed in customer deliverables
- Lower level requirements can be traced to a valid source
- Requirements are implemented
- Selected requirements are verified and validated
- Relationships to other entities such as intermediate and final work products may:
  - Include dependencies in which a change in one requirement can impact other requirements
  - Aid in design and in evaluating the impact of changes
  - Affect evaluating the impact of requirements changes
  - Support anticipated scope changes
- Design and other documentation reflect requirements
- Test plans and test cases address the requirements

Traceability is particularly helpful when assessing the impact of requirements changes on work activities and projects. Bidirectional traceability is not always automated. It can be done manually using spreadsheets, databases, and other common tools. Bidirectional traceability needs to be implemented in concert with lifecycle activities. If it is left to the end of the project, it will be a costly and error prone effort.

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Example Activities

Example Activities	Further Explanation
Develop, record, and keep updated bidirectional requirements traceability.	Trace requirements from their source through intervening work products to the customer deliverable

Example Work Products

Example Work Products	Further Explanation
Records of bidirectional requirements traceability	

Context Specific

Safety

Context Tag:	CMMI-SAF
Context:	Use processes to incorporate safety considerations as an integral part of the solution, work, project, and organization.

Ensure requirements incorporate safety requirements, as based on the results of hazard identification, hazard analysis, and risk assessment. Ensure safety requirements are traceable to the related hazard.

Services

Context Tag:	CMMI-SVC
Context:	Use processes to deliver, manage, and improve services to meet customer needs.

In a service environment, trace stakeholder requirements to:

- Elements of the delivered service and supporting service system developed from those requirements
- Service levels
- Other requirements derived from stakeholder requirements

Elements of the delivered service and supporting service system should be traceable back to the stakeholder requirements they meet.

Maintain traceability for work products and components, such as the service system architecture, service system components, development iterations, or increments, functions, interfaces or connections, objects, people, processes, and other work products.

A traceability matrix might have the list of stakeholder requirements and derived requirements on one axis. The other axis might list components of the service system, including people and consumables. The intersections of the rows and columns would indicate where a requirement applies to the parts of the service system.

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Supplier Management

Context Tag:	CMMI-SPM
Context:	Use processes to identify, select, and manage suppliers and their agreements.

The acquirer maintains overall bidirectional traceability between customer requirements and the solution. The supplier maintains bidirectional traceability between the solution, the solution components, and the requirements defined in the supplier agreement. The acquirer verifies that traceability. The supplier also maintains traceability from contractual requirements to derived or additional requirements.

RDM 2.6

Required Practice Information

Practice Statement

Ensure that plans and activities or work products remain consistent with requirements.

Value

Minimizes rework by eliminating inconsistencies between requirements and related artifacts.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

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Example Activities

Example Activities	Further Explanation
Review plans, activities, and projects for consistency with requirements and changes made to them.	Maintaining bidirectional traceability is critical to maintaining consistency between requirements, work products, and plans.
Record inconsistencies and their sources.	Identify any changes that should be made to plans and projects resulting from changes to the requirements.
Initiate and record any necessary corrective actions and communicate results to affected stakeholders.	

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Example Work Products

Example Work Products	Further Explanation
Records of inconsistencies between requirements, plans, and work products	
Corrective actions	

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Level 3

RDM 3.1

Required Practice Information

Practice Statement

Develop and keep requirements updated for the solution and its components.

Value

Ensures the built solutions meet the customers' needs and expectations in a consistent way across the organization.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

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Example Activities

Example Activities	Further Explanation
Develop, record, and keep updated requirements in technical terms necessary for solution and solution component design.	
Derive, record, and keep updated requirements that result from solution selections and design decisions.	
Record and keep updated bidirectional traceability.	
Review requirements for potential impact to architecture and design.	
Record and keep updated prioritization of requirements.	
Record and keep updated nontechnical requirements.	
Identify, record, and keep updated requirements for external and internal interfaces or connections.	

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Example Work Products

Example Work Products	Further Explanation
Requirements	May include: • Solution requirements • Architectural requirements – specify or constrain the relationships among components • Solution component requirements • Derived requirements – with bidirectional traceability to source requirements • Requirement allocations – for bidirectional traceability • Internal and external interface or connection requirements

Context Specific

Development

Context Tag:	CMMI-DEV
Context:	Use processes to develop quality products and services to meet the needs of customers and end users.

Selection of a technology brings with it additional requirements. For instance, use of electronics requires additional technology-specific requirements such as electromagnetic interference limits.

Services

Context Tag:	CMMI-SVC
Context:	Use processes to deliver, manage, and improve services to meet customer needs.

Analyze stakeholder requirements while developing the operational concept to derive more detailed and precise sets of requirements called derived requirements. These requirements address all aspects of the service system associated with service delivery, including work products, services, processes, consumables, customer resources, other resources, warranty costs, service incentives, and the functionality and quality attribute needs of affected stakeholders.

In some service contexts, derived requirements can be as simple as identifying and quantifying required resources. For complex service systems with many types of components and interfaces, iteratively refine the initial requirements into lower level sets of more detailed requirements that can be allocated to service system components as the preferred solution is refined.

Develop the functionality and quality attribute requirements for the service system through this analysis and through refinement, derivation, and allocation activities.

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RDM 3.2

Required Practice Information

Practice Statement

Develop operational concepts and scenarios.

Value

Enables customers to understand, confirm, and commit to how their requirements will be met.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

A scenario describes a sequence of events that:

- Includes user and product interactions
- May occur in the development, use, or sustainment of the product

An operational concept provides:

- A view of the system or product from the user perspective
- A context for developing or evaluating a set of scenarios

Operational concepts and scenarios together demonstrate what the requirements are trying to accomplish. For example, the operational concept for a satellite-based communications product is quite different from one based on landlines. The scenario would cover the steps in the communication process.

Example Activities

Example Activities	Further Explanation
Develop, record, and keep updated operational concepts and scenarios.	Identify and develop scenarios, consistent with the level of detail in the stakeholder needs, expectations, and constraints. Ensure scenarios address quality attribute, functions, or other factors. Iteratively refine operational concept and scenarios to include more detail as decisions are made and as lower level requirements are developed; for example, describe interactions among the: • Solution • End users • Environment • Boundaries and constraints

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Example Activities	Further Explanation
Review operational concepts and scenarios with affected stakeholders to refine and discover requirements.	

Example Work Products

Example Work Products	Further Explanation
Operational concepts and scenarios	

RDM 3.3

Required Practice Information

Practice Statement

Allocate the requirements to be implemented.

Value

Increases customer satisfaction by delivering a complete solution that meets requirements.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

The architecture provides the basis for allocating requirements to components. A higher-level requirement can be allocated to more than one component.

Example Activities

Example Activities	Further Explanation
Allocate, record, and keep updated requirements.	Requirements are typically allocated to: • Logical design – As the operational concept evolves, allocate requirements to logical entities, e.g., functions, processes, that help relate the requirements to the operational concept. These logical entities serve to organize the requirements and assist in the synthesis of the technical solution. • Physical design – As the technical solution is selected or developed, allocate requirements to solution components. • Components

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Example Activities	Further Explanation
	• Interfaces or connections • Delivery increments
Record and keep updated relationships among allocated requirements.	Relationships include dependencies in which a change in one requirement can affect other requirements.
Review requirements allocations and relationships with affected stakeholders.	

Example Work Products

Example Work Products	Further Explanation
Requirement allocations	

Context Specific

Supplier Management

Context Tag:	CMMI-SPM
Context:	Use processes to identify, select, and manage suppliers and their agreements.

Allocate contractual requirements, as appropriate, to supplier deliverables. Record the requirements for each supplier deliverable. In some cases, technical requirements are allocated to third-party solutions that the supplier uses, e.g., COTS products.

Example acquirer activities include:

- Allocating requirements to suppliers
- Allocating requirements to supplier deliverables
- Allocating design constraints and record relationships among allocated requirements and design constraints
- Developing an approach to address requirements shared among multiple stakeholders, e.g., the acquirer, multiple suppliers, customers, end users

RDM 3.4

Required Practice Information

Practice Statement

Identify, develop, and keep updated interface or connection requirements.

Value

Reduces rework and risk due to incompatible internal and external interfaces or connections.

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Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Identify interfaces or connections between functions, objects, or other logical entities. Interfaces or connections can be internal or external to a solution. Define interface or connection requirements between solutions or solution components identified in the architecture. Interfaces or connections are controlled as part of solution integration. Ensure interface or connection requirements are reviewed by affected stakeholders. This is typically done within the context of a control group of stakeholders.

Example Activities

Example Activities	Further Explanation
Identify, record, and keep updated internal and external interface or connection requirements.	As the design progresses, the architecture will be altered by technical solution processes, developing new interfaces or connections between internal components and components external to the solution. Management of the interfaces or connections may include: • Maintaining consistency of the interface or connection • Complying with architectural decisions and constraints • Resolving conflict, non-compliances, and change issues
Review interface or connection requirements for coverage and completeness with affected stakeholders, and record results.	Review the interface or connection descriptions with affected stakeholders to: • Avoid misinterpretations • Reduce delays • Prevent the development of interfaces or connections that do not work properly

Example Work Products

Example Work Products	Further Explanation
Interface or connection requirements	May include: • Categories with lists of interfaces or connections • Requirements defined for each set of solution components
Interface or connection requirements review results	
Action items for updating interface or connection requirements	

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Example Work Products	Further Explanation
Updated interface or connection requirements	

Related Practice Areas

Product Integration (PI)

Technical Solution (TS)

Context Specific

Development

Context Tag:	CMMI-DEV
Context:	Use processes to develop quality products and services to meet the needs of customers and end users.

Define requirements for interfaces in terms such as:

- Software
  - Origination
  - Destination
  - Stimulus
  - Data characteristics
  - Communication interfaces, e.g., protocols such as HTTP, FTP
  - User interfaces, e.g., screen layout, font, buttons
- Hardware
  - Electrical
  - Mechanical
  - Hydraulic
  - Physical connections
  - Power characteristics
  - Radiation
  - Thermal
  - Safety
  - Security

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RDM 3.5

Required Practice Information

Practice Statement

Ensure that requirements are necessary and sufficient.

Value

Avoids rework by only delivering necessary solutions.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

A necessary requirement is one that must be met for the solution to function as intended. Sufficiency typically deals with the set of requirements being complete enough for the solution to function as intended.

Requirements analysis helps determine if requirements:

- Are all necessary
- Are missing
- Are consistent with each other
- Can be implemented as defined
- Can be verified and validated
- Conflicts are removed

Example Activities

Example Activities	Further Explanation
Perform a requirements analysis to determine if requirements are necessary and sufficient.	Analyze requirements to ensure they are: • Relevant • Complete • Feasible • Maintainable • Not in conflict Analyze operational concepts and scenarios to refine customer needs, constraints, and interfaces or connections, and to discover new requirements. This analysis can result in more detailed operational concepts and scenarios and supports deriving new requirements.

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Example Activities	Further Explanation
Review analysis results with stakeholders.	
Update requirements based on review results.	

Example Work Products

Example Work Products	Further Explanation
Requirements analysis results	May include reviews of: • Requirements defects • Activity diagrams and use cases • Logical or functional designs with allocated requirements • Proposed requirements changes
Updated requirements	

RDM 3.6

Required Practice Information

Practice Statement

Balance stakeholder needs and constraints.

Value

Increases stakeholder satisfaction while addressing conflicting requirements and constraints.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Ensure an understanding and agreement between the project and the customer about what is delivered given resource constraints. This typically involves negotiating benefits and tradeoffs with customers and other stakeholders. When costs and issues outweigh the benefits, consult affected stakeholders to determine what changes may be needed.

Stakeholder needs and constraints can address:

- Cost
- Schedule
- Performance

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- Functionality
- Quality
- Priorities
- Resources
- Reusable components
- Maintainability
- Risk

Analyze requirements to determine whether they reflect an appropriate balance among cost, schedule, performance, quality, customer needs, and other factors of interest to affected stakeholders. Models and simulations can be used to estimate the impact that requirements will have on these factors. Involve stakeholders in the analysis of impacts and issues from different phases of the solution's lifecycle. If the issues are considered unacceptable, revise or reprioritize the requirements to improve the balance of cost, schedule, performance, and quality.

Example Activities

Example Activities	Further Explanation
Develop and keep updated a definition of required functionality and quality attributes.	This definition can include descriptions, breakdowns, and categorizations of the functions. A clear understanding of the quality attributes and their importance based on business objectives or needs is essential to the design process.
Analyze requirements to ensure that they balance stakeholder needs and constraints.	Analysis includes use of: • Models • Simulations • Prototypes • A risk assessment on the requirements and definition of required functionality, performance, and quality attributes • Assessment of the requirements on cost, schedule, functionality, and quality
Review, analyze and negotiate requirements tradeoffs with customers and stakeholders.	
Record and keep updated proposed requirements changes and communicate with affected stakeholders.	
Validate requirements to ensure the resulting service system will perform as intended in the end user's environment.	

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Example Work Products

Example Work Products	Further Explanation
Operational concepts and scenarios, use cases, activity diagrams, and user stories	This may include: • Required functionality and quality attributes • Architectural concepts • Operational and maintenance concepts, e.g., installation, training, disposal
Analysis methods and results	Includes: • Identification of where costs, schedule, performance, quality, and other factors exceed constraints • Assessment of risks related to requirements
Proposed requirements changes	

Related Practice Areas

Risk and Opportunity Management (RSK)

Context Specific

Services

Context Tag:	CMMI-SVC
Context:	Use processes to deliver, manage, and improve services to meet customer needs.

Analyze requirements and define required service system functionality and quality attributes to balance the stakeholders' needs, expectations, constraints, and interfaces or connections. Depending on the service delivery context, consider factors such as feasibility, business objectives and needs, cost constraints, end user types, potential market size, and procurement strategy. Determine the parameters used to evaluate the effectiveness of service delivery based on customer and end user input and the preliminary service delivery concept.

Supplier Management

Context Tag:	CMMI-SPM
Context:	Use processes to identify, select, and manage suppliers and their agreements.

Perform a cost benefit analysis to assess trade-offs between requirements and the effect on the overall acquisition strategy.

This analysis often focuses on evaluating requirements that address architecturally significant quality attributes. For example, a combination of tight response time requirements and high reliability requirements could be expensive to implement. Impact analysis provides the insight for the acquirer to select a more cost effective solution to balance cost, schedule, and performance against risk and opportunity.

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RDM 3.7

Required Practice Information

Practice Statement

Validate requirements to ensure the resulting solution will perform as intended in the target environment.

Value

Avoids rework cost and increases satisfaction by delivering a solution that meets customer expectations and needs.

Additional Required Information

This section left blank for future content.

Explanatory Practice Information

Additional Explanatory Information

Perform requirements validation with affected stakeholders throughout the solution lifecycle to confirm that the requirements are necessary and sufficient in the target environment.

Example Activities

Example Activities	Further Explanation
Identify and select validation techniques.	To broaden and deepen understanding of requirements, use multiple validation techniques such as: • Functional analysis • Simulation • Prototypes • Demonstrations • Tests • Reviews or walk-throughs
Validate the requirements using selected techniques and record results.	
Review and communicate validation results with stakeholders.	
Update requirements.	

Example Work Products

Example Work Products	Further Explanation
Selected validation techniques	

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Example Work Products	Further Explanation
Record of validation results	
Updated requirements	

Related Practice Areas

Verification and Validation (VV)

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